

Phosphorus

P (P I). Strongest classified transitions below 2 μm (NIST): 23 levels, 50 lines.

P⁺ ionisation limit 84580.83 cm⁻¹ = 10.48669 eV (118.230 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $|⟨J||er||J'⟩|$ in ea₀
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, *n* = 70.
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).

